

DIAC 2024 Group18

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I. INTRODUCTION

A Telescopic Operational Transconductance Amplifier (OTA) is a type of a differential amplifier that is commonly used in analogue IC-design, especially for high-speed and low-power applications. Its architecture is characterized by a "telescopic" arrangement of transistors (as shown in Figure 1), where the signal path extends through stacked transistors, optimizing gain while keeping power consumption and output swing limitations in order. This design allows telescopic OTAs to achieve high gain or high bandwidth (*gain-bandwidth trade-off*) with relatively low noise levels compared to other OTA configurations, such as folded-cascode OTAs. However, the telescopic configuration comes with some trade-offs, such as a reduced output voltage swing, making it suitable for specific environments where these parameters are acceptable.

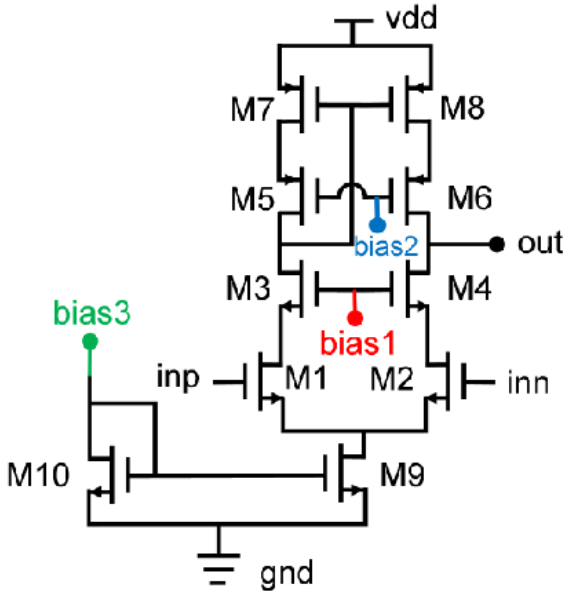


Figure 1: Telescopic OTA Schematic.

This article covers the basic steps in the design process of a telescopic OTA, and the results follow as well.

II. FIRST ORDER APPROXIMATIONS

To calculate the DC operating point for the telescopic OTA design, the maximum current was set at 80 μA (the given maximum current through transistor M9 is $I_{M9\text{max}}=85 \mu\text{A}$) aiming for a DC output voltage of 0.9 V. The requirement for this was to carefully select transistor operating points and ensure that all transistors would remain in good operating area (saturation). For the overdrive voltages in the calculations, a commonly used value of the voltage overdrive in 180 nm CMOS technology was chosen - 0.2 V. With the given current and overdrive voltages, it is possible to calculate the transconductance with the following equation:

$$g_{m1,2} = \frac{2 * I_{DS}}{(V_{GS} - V_{TH})} = 400 \frac{\mu\text{A}}{\text{V}} \quad (1)$$

To set the bias voltages for the transistors, the chosen overdrive voltages were used, as well as threshold voltages and the drain-to-source voltages equal to the average voltage drops considering that DC output voltage remains 0.9 V (for nMOS transistors 0.3V and for pMOS transistors -0.45V).

$$V_{Dsat} = V_{GS} - V_{TH} = V_{OD} = 0.2 \text{ V} \quad (2)$$

$$V_{B1} = 2V_{DS} + V_{TH} + V_{OD} = 1.435 \text{ V} \quad (3)$$

$$V_{B2} = V_{DD} + V_{DS} + V_{TH} + V_{OD} = 512 \text{ mV} \quad (4)$$

The W/L ratio is calculated from the expressions (5) and (6) which were derived from the transistors' currents equations in saturation mode:

$$\left(\frac{W}{L}\right)_n = \frac{I_{DS}}{\beta_n V_{OD}^2} = 11.49 \quad (5)$$

$$\left(\frac{W}{L}\right)_p = \frac{I_{DS}}{\beta_p V_{OD}^2} = 55.56 \quad (6)$$

Using the expression (7), the output resistance was obtained, and used in expression (8) to calculate the ratio of the lengths of the nMOS and pMOS transistors. By iterating the nMOS length, the chosen length was 4 μm .

$$A_V = g_{m1,2} * R_{out} \quad (7)$$

$$R_{out} = r_{ds1,2} g_{m3,4} r_{ds3,4} || r_{ds7,8} g_{m5,6} r_{ds5,6} \quad (8)$$

III. DC OPERATING POINT

After the first order approximations and hand calculations, a schematic of the telescopic OTA was made in *Cadence Virtuoso* (similar to the schematic shown in Figure 1) and the calculated parameters (transistor dimensions, biasing voltages and current) were inserted to begin different analyses of the amplifier. The first step is to check whether all the transistors are in sufficient operating area (saturation) and whether their operating points are optimal. This could happen to be an iterative job, as some values may have not been calculated correctly (e.g. parasitic capacitances were neglected, hence transconductance was not entirely precisely calculated etc.) - the need for corrections occurs. For example, the DC output voltage, after running the DC analysis, was 970 mV. To get to the given 900 mV the width of the pMOS transistors and the bias voltage of transistors M3 and M4 were iteratively shorted. Without degrading the output signal, the final DC output voltage is 906 mV.

The results of the DC analysis are shown in Table 1 – it contains the width and the length of each transistor, their ratios, as well as the values of their transconductances, output resistances, drain-to-source voltages, gate-to-source voltages and overdrive voltages. The final values of the biasing voltages are $V_{b1}=1.285 \text{ V}$, $V_{b2}=0.552 \text{ V}$ and the final biasing current is $I_B=21.25 \mu\text{A}$, which results in a mirrored current of $I_{M9}=83.57 \mu\text{A}$ (the ideal current mirror ratio used in this case is 1:4).

<i>n</i> MOS transistors	W [μm]	L [μm]	W/L	V_{OD} [mV]	V_{GS} [mV]	V_{DS} [mV]	g_m [$\mu\text{A}/\text{V}$]	r_{ds} [$k\Omega$]
<i>M1, M2</i>	60	4	15	139.3	792.6	265.4	445.8	454.96
<i>M3, M4</i>	48.8	4	12.2	152.1	812.1	437.8	410.9	1332.8
<i>M9</i>	146.4	6	24.4	153.2	809.2	207.4	812	66.58
<i>M10</i>	36.6	6	6.1	153.1	809.2	809.2	207.5	6426.7
<i>p</i> MOS transistors								
<i>M5, M6</i>	156.12	6	26.02	-212.5	-934.9	-576.2	273.7	3361.3
<i>M7, M8</i>	218.58	6	36.428	-182.9	-889.3	-313.1	317.8	408.16

Table 1: DC operating point results for all transistors.

IV. AC ANALYSIS

To assure desired gain and the gain-bandwidth product (GBW in latter text), AC analyses needed to be made. The focus was mainly on the gain characteristic of the amplifier, as the low frequency gain (A_{v0} in latter text) and the GBW could have been derived from this characteristic. The characteristic is shown in Figure 2, and the results are as follows:

$$|A_{v0}| = 96.55 \text{ dB} \quad (9)$$

$$GBW = 2.11 \text{ MHz} \quad (10)$$

N. B. The results were obtained by using markers in *Cadence Virtuoso*.

It is important to understand that, to get the desired specifications, more corrections of the values calculated in the first order approximation may have been made. Considering the trade-offs, such as the gain-bandwidth trade-off etc., the crucial part of the design was determining the optimal values of performance parameters which yield a desired outcome.

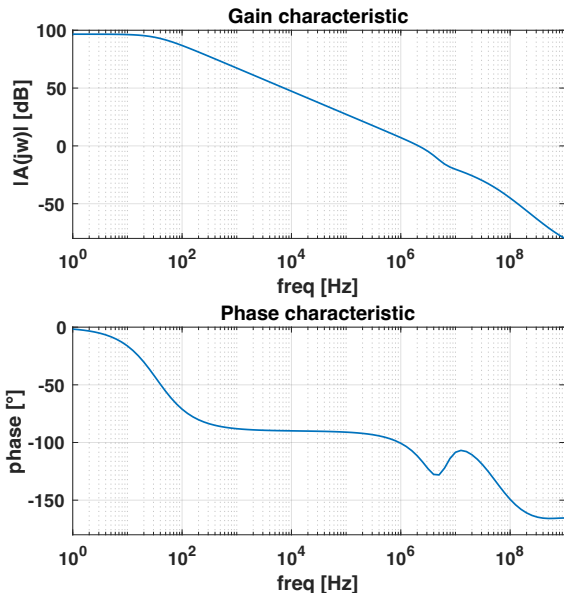


Figure 2: Gain and Phase Characteristics of the OTA.

V. TRANSIENT ANALYSIS

After acquiring the needed parameters' values, it is high time to test the performance of the amplifier. This may be seen when running the transient analysis – a sinusoidal wave was applied as the input signal, then the input and the output signal were plotted, which is shown in Figures 3 to 5. As was required, the voltage swing is the desired one, but the output signal is distorted. The distortion may have occurred due to inappropriate biasing or transistors' dimensions – even though the parameters (mostly biasing voltages) have been changed iteratively in this step as well, a better outcome could not have been obtained.

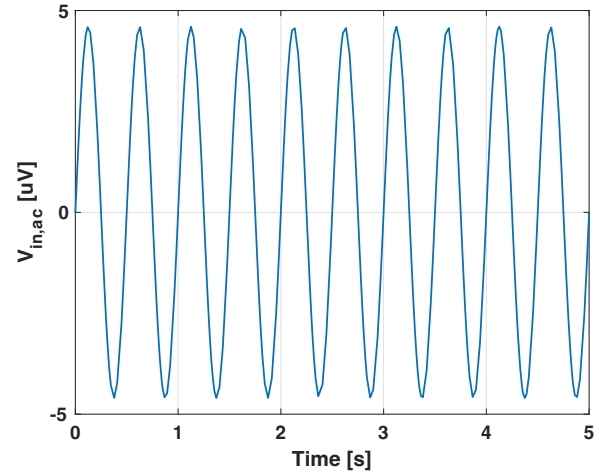


Figure 3: Transient analysis results – input signal.

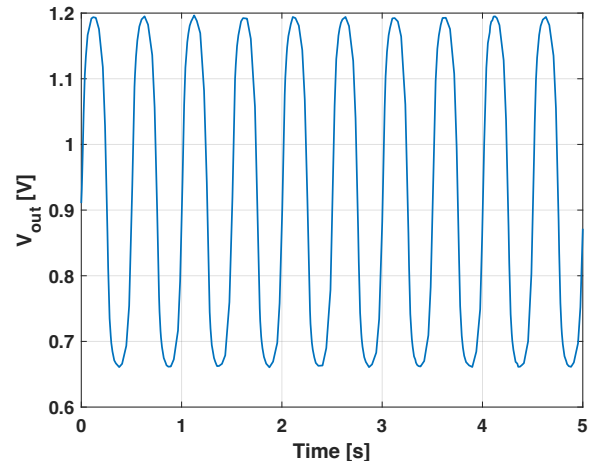


Figure 4: Transient analysis results – output signal.

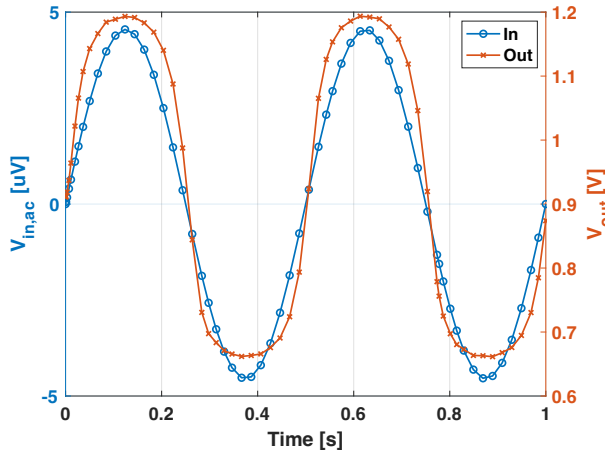


Figure 5: Transient analysis results – input and output signal, zoomed in.

VI. LAYOUT DESIGN

After having done all the necessary analyses, transistor parameters are now fixed, and one may proceed with the design process. Yet another crucial step is the layout design which is later needed for a successful fabrication of the desired product. Layout had also been made in *Cadence Virtuoso* – all the transistors were imported and connected as is described in the schematic. Bigger transistors were split into more smaller ones to reduce the parasitics, and the dummy transistors were introduced for more matched and symmetric network. Indeed, an optimal symmetry is obligatory for a proper layout design of the circuit.

While designing a layout, every now and then the designer should conduct Design Rule Check (in latter text DRC) which notifies one about the possible errors made in the design regarding the nets' connectivity, spacing etc. Other than DRC, an important aspect of this part is conducting the Layout Versus Schematic (in latter text LVS) analysis which notifies one whether the nets and pins in the layout are connected the same way as in the schematic. When the layout for the OTA is finished, it needs to be put into the *padding*, the pins should be connected to the *padding* and, if everything is done correctly, DRC and LVS should be clean in the end.

After successfully performing DRC and LVS, the layout design is ready to be exported and sent to the fabrication. The layout of the telescopic OTA put into the *padding* is shown in the Figure 6.

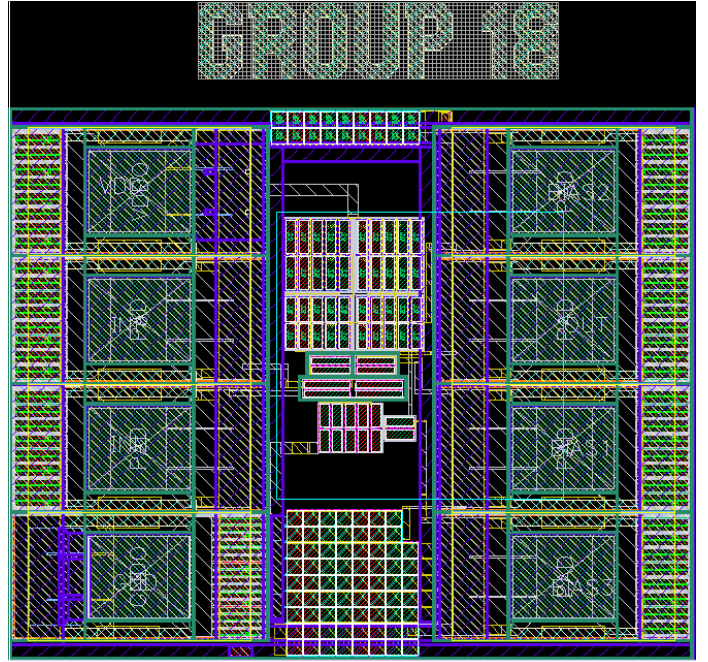


Figure 6: The layout of the telescopic OTA.

VII. CONCLUSION

Designing an analogue integrated circuit is a process consisting of a few crucial steps, and every step needs to be carefully made to get an optimal and efficient product. The most important parameters need to be chosen wisely – biasing, the components' dimensions, power consumption etc. all need to yield a useful and feasible solution. Nevertheless, designers need to be aware of potential trade-offs and consequences their choices have, and they need to prepare themselves that *a free lunch* indeed does not exist.